

STAT/MA 41600
In-Class Problem Set #2: August 28, 2026
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Problem Set 2 Answers

1a. There are 2^{100} equally-likely results, and $\binom{100}{50} = \frac{100!}{50!50!}$ of them consist of 50 heads and 50 tails. So the probability is $\binom{100}{50}/2^{100} = 0.07959$.

1b. The probability that the number of heads and tails are different is $1 - \binom{100}{50}/2^{100}$. In such a case, by symmetry, there are strictly more heads than tails half the time. So the probability that there are strictly more heads than tails is $1/2 - \binom{100}{50}/2^{101}$.

2. There are 36 equally-likely outcomes, 23 of which are between 4 and 8 (inclusive). So the probability is $23/36$.

2b. There are 36 equally-likely outcomes; for 16 of these outcomes, we have $|X - Y| \leq 1$. So the probability is $16/36$.

3a. There are $6^3 = 216$ equally-likely outcomes, and $5^3 = 125$ of these are bear families that do not contain any red bears. So the probability is $125/216$.

3b. There are $216 - 6 = 210$ bear families that are *not* primary. So the probability is $210/216$.

3c. There are six possible ways that all three bears have the same color. So the probability is $6/216$.

3d. There are $(6)(5)(4) = 120$ ways that the three bears have different colors. So the probability is $120/216$.

4. There are $3^5 = 243$ equally-likely outcomes for the weekly weather forecasts. For each value of k from 0 to 5 (inclusive), there are $\binom{5}{k}2^{5-k}$ such weekly weather forecasts, so the probability is $\binom{5}{k}2^{5-k}/3^5$.

5. The probability is $\sum_{x=2}^{\infty} \frac{3^x e^{-3}}{x!} = (\sum_{x=0}^{\infty} \frac{3^x e^{-3}}{x!}) - e^{-3} - 3e^{-3} = 1 - 4e^{-3}$.

Mind bender (not required). Returning to the situation in Question 1, the probability of exactly k heads is $\binom{100}{k}(1/2)^k(1/2)^{100-k}$. By the binomial theorem, we have

$$\sum_{\text{all } k} \binom{100}{k} (1/2)^k (1/2)^{100-k} = (1/2 + 1/2)^{100} = 1$$

and

$$\sum_{\text{all } k} \binom{100}{k} (-1/2)^k (1/2)^{100-k} = (-1/2 + 1/2)^{100} = 0$$

Adding these two equations, we notice that the contributions from the odd values of k will cancel on the left-hand-side, and the contributions from the even values of k are repeated in both equations, so we get

$$2 \sum_{\text{even } k} \binom{100}{k} (1/2)^k (1/2)^{100-k} = 1.$$

Dividing by 2 throughout, we get: $\sum_{\text{even } k} \binom{100}{k} (1/2)^k (1/2)^{100-k} = 1/2$.